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Pioneering Deployment of Advanced E-Line Artificial Lift Technology for First Time in UAE

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Abstract

This paper evaluates the performance of novel advanced artificial lift technology, deployed for the first time in the UAE, across four oil-producing wells from two different ADNOC Onshore assets in 2024. It aims to highlight the benefits in reviving ceased wells, enhancing efficiency, reducing energy use, and improving safety and sustainability. The study examines its effectiveness in challenging conditions and its potential to transform artificial lift strategies for optimized production with reduced carbon footprint.

The methodology focused on a detailed evaluation of the advanced artificial lift technology through UAE's first field trials in 2024, conducted on four oil-producing wells in two fields of ADNOC Onshore. Wells that had ceased production and previously relied on nitrogen lift, were selected to deploy the permanent magnet motor-driven system via a through-tubing method. Critical parameters such as production rates, energy consumption, and system reliability were tracked, while also assessing safety improvements and emission reductions. Data on uptime and performance in slim-bore and deviated wells was collected and analyzed to determine the technology's effectiveness and potential for wider use in complex reservoirs.

The 2024 field trials of the advanced artificial lift technology across four oil-producing wells in two UAE fields produced encouraging outcomes. In the first field, both wells maintained consistent natural flow post-deployment, confirming the system's effectiveness in reviving ceased wells without relying on nitrogen lift. In the second field, both wells flowed successfully while the technology was active; however, after the tool was removed following the restoration of flow, the wells ceased production due to high water cut. The system delivered notable improvements in efficiency, cutting energy use and carbon emissions compared to conventional coiled tubing methods. Its compact design and through-tubing deployment eliminated the need for rig interventions, reducing lifecycle costs and enhancing safety by replacing coiled tubing operations. Technology also enabled production from slim-bore wells, opening up new opportunities in challenging reservoirs. It performed reliably in deviated and horizontal wells, showcasing its versatility. In conclusion, this technology presents a game-changing approach to artificial lift, offering a cost-effective, efficient, and

eco-friendly solution for reactivating dormant wells. It supports the UAE's sustainability objectives and holds significant potential for wider adoption, particularly in complex reservoir settings.

This paper unveils a groundbreaking 2024 UAE trial of advanced artificial lift technology, boldly challenging the reliance on conventional techniques such as nitrogen lift and rig interventions. Unlike traditional methods, this cutting-edge system—driven by permanent magnet motor technology with through-tubing deployment—slashes costs, boosts safety, and drastically cuts energy use while accessing slim-bore wells in complex reservoirs. Publishing this work is vital to spotlight a revolutionary, sustainable shift that upends conventional practices and propels the petroleum industry forward with transformative innovation.

Introduction

Reactivating ceased or low-performing oil and gas wells is a complex undertaking, particularly in mature fields[1] and challenging reservoir conditions. These wells often exhibit formation damage, scale buildup, high water cut, and compromised integrity, which limit the effectiveness of conventional artificial lift methods. Coiled tubing (CT) interventions—commonly used for nitrogen kick-off—are energy-intensive and environmentally taxing due to the deployment of multiple diesel-powered units such as power packs, triplex pumps, and nitrogen systems. CT operations also carry significant safety risks, involving high-pressure pumping, hazardous chemicals, and extensive surface equipment that demand strict HSE protocols and specialized protective gears. The need for continuous 24-hour execution further elevates exposure and logistical complexity. Mechanically, CT faces limitations in slim-bore and deviated wells, where access constraints can lead to stuck pipe or coil tubing string failure under stress. Well control remains a critical concern, requiring precise pressure management to prevent blowouts or formation damage. These drawbacks—high carbon footprint, elevated safety risks, and operational inefficiencies—highlight the need for innovative, rigless artificial lift technologies that offer safer, more sustainable, and cost-effective alternatives for well activation.

Problem Statement

Conventional well activation techniques, particularly those involving coiled tubing (CT) and nitrogen lift —have long been deployed to restore flow in ceased or underperforming wells. However, these methods present a range of limitations that increasingly conflict with modern oilfield standards for efficiency, safety, and sustainability. CT-based nitrogen lifting is inherently resource-intensive, requiring multiple diesel-powered units such as triplex pumps, nitrogen systems, and hydraulic power packs. This setup contributes to a substantial carbon footprint and elevated fuel consumption, making it a less viable option in the context of decarbonization and emission reduction goals.

From a safety and well control perspective, CT operations are classified as high-risk activities. This is primarily due to the fact that they involve working on a live well—where the CT string is inside the tubing while the well is actively flowing. This scenario introduces significant hazards, including the potential for uncontrolled flow, pressure surges, and blowouts. The handling of high-pressure systems and cryogenic fluids like liquid nitrogen further compounds the risk, necessitating rigorous HSE protocols, specialized PPE, breathing apparatus, and continuous monitoring. The complexity of surface equipment and the need for 24-hour execution elevate exposure levels and demand highly trained personnel to manage operational contingencies.

Technically, CT interventions face mechanical vulnerabilities that limit their reliability in complex well architectures. Slim-bore and highly deviated completions pose access constraints that increase the likelihood of stuck pipe, tubing collapse, and fatigue-induced failures. In extended-reach or horizontal wells, depth limitations due to buckling and drag forces restrict CT's ability to reach target zones effectively. Moreover, in wells with high water cut, nitrogen injection may only offer temporary flow restoration, with production often declining once the lift is discontinued.

Operationally, CT activation demands precise pressure management to maintain well control and avoid formation damage. The need for continuous nitrogen supply and real-time diagnostics adds to logistical complexity, especially in remote or infrastructure-constrained locations. Mobilizing heavy surface equipment, coordinating specialized crews, and ensuring uninterrupted fluid delivery all contribute to elevated costs and extended timelines.

These cumulative challenges—ranging from environmental impact and safety exposure to mechanical limitations and operational inefficiencies—underscore the urgent need for next-generation artificial lift technologies. Rigless systems with compact designs, lower energy requirements, and enhanced adaptability to complex reservoir conditions offer a promising alternative. Such innovations not only reduce lifecycle costs and carbon emissions but also improve safety and reliability, aligning with the industry's evolving priorities for sustainable and efficient well activation strategies.

Proposed Workflow

To ensure safe, efficient, and sustainable reactivation of ceased wells, a structured and technically sound workflow is essential—especially when deploying novel artificial lift technologies in complex reservoir environments. The following proposed workflow outlines the key stages for rigless deployment of advanced artificial lift systems, optimized for slim-bore, deviated, and high water cut wells:

Proposed Workflow for Rigless Artificial Lift Deployment

Candidate Well Selection.

- Identify ceased or low-performing wells with prior reliance on CT-based nitrogen activation.
- Review historical production data, water cut trends, and well integrity status.

Pre-Deployment Diagnostics.

- Conduct production logging and PLT surveys to assess inflow profiles and identify damage zones (optional).
- Evaluate wellbore geometry, completion type, and accessibility for through-tubing deployment.
- Perform well integrity checks including pressure tests, barrier validation, and H_2S risk assessment.

Technology Configuration.

- Select appropriate artificial lift system[2].
- Customize tool length, power rating, and deployment accessories based on well depth and deviation.
- Ensure compatibility with existing completion hardware and tubing ID.

Deployment Planning.

- Develop a rigless intervention plan using e-line conveyance.
- Schedule surface equipment mobilization and coordinate logistics.

Field Execution.

- Deploy the artificial lift tool through tubing without rig intervention.
- Monitor real-time parameters such as current draw, motor speed, and downhole temperature.
- Initiate flowback and stabilize production, ensuring safe handling of live well conditions.

Post-Deployment Evaluation.

- Track production rates, energy consumption, uptime, and system reliability.
- Compare post-deployment performance against baseline (pre-activation) metrics.
- Analyze efficiency gains, carbon footprint reduction, and safety improvements.
- Document lessons learned, including tool behavior in slim-bore and deviated wells.
- Recommend optimization strategies for future deployments in similar reservoir settings.

Case Study: Well A, Field X (Oil Producer with horizontal open hole completion):

The well was initially drilled and completed as a vertical oil producer. It was later worked over and recompleted as a dual producer across two reservoir intervals. Following downhole communication issues, the well was sidetracked and re-entered with a horizontal completion using a single tubing string. A subsequent workover was performed to replace a malfunctioning safety valve, and the well was re-equipped with a single completion string incorporating a surface-controlled safety system. After reactivation, the well resumed production. However, the productivity index declined significantly, prompting plans for stimulation and nitrogen kick-off to enhance flow performance. Unfortunately PLT results were not available for investigation but multiple production tests carried out over years confirmed consistent water cut of 40%, GOR approximately 1000 scf/bbl and liquid rate in the order of thousands bbl/day.

Operational history indicates that the well is sensitive to flow interruptions, often requiring nitrogen kick-off following interventions or wellhead maintenance. Each previous kick-off event typically involved the pumping of approximately 4,000 US gallons of nitrogen using CT and the activity spanned a minimum of six days at a time (including mobilization, rig up, pumping and rig down), underscoring the logistical and resource demands associated with maintaining stable production from this asset.

Slick line was used to set the SIM Plug Packer at 3700 ft (vertical section), followed by a rigless electric wireline run with the novel ESP BHA to latch onto the packer. After confirming the latch, the control room lined up the well for flow. The novel pump was kickstarted at 6000 rpm and flow was confirmed; the pump was run for about 30 minutes to monitor flow, then was stopped intermittently only to observe continued downhole parameters. The pump ran for approximately 2 hours. During this time well showed stable downhole and surface pressures. After a 30-minute intentional shut-in to monitor parameters, downhole pressure remained stable. The electric wireline and BHA were pulled out of hole, and the well resumed healthy flow. It was monitored for 7 days and maintained consistent production throughout.

As anticipated, there was no change in PI; however, this successful well activation demonstrated that the novel pump—featuring a reduced carbon footprint (utilizing only a crane, e-line unit, and power pack)—achieved equivalent results within 2 hours. In contrast, conventional coil tubing operations, which involve a significantly larger carbon footprint (including crane, coil tubing, triplex pumps, nitrogen pump, power pack, and centrifugal pump), would require 16 hours to deliver the same output by pumping 4,000 US gallons of liquid nitrogen.

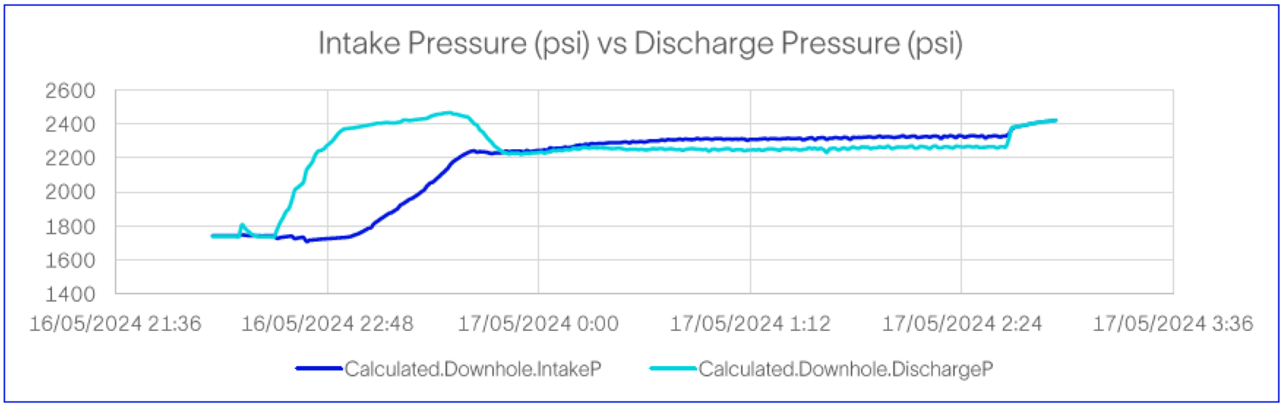

Figure 1—Novel Pump Intake & Discharge Pressures for Well A

Case Study: Well B, Field X (Oil Producer with horizontal cased hole completion & GLMs)

The well was drilled and completed as a single horizontal oil producer (cased hole) and is equipped with 4 SPMs in 3 ½" completion string. The well has history of high water (upto 50%) and scale production. The latest slickline report shows no accessibility till deepest nipple with biggest size gauge cutter.

Though the well is equipped with 4 gas lift mandrels but since the gas lift line has not been laid yet completely in this field, the well can not be supported with gas lift yet. Just as previous candidate this well is also sensitive and requires nitrogen lifting whenever it ceases to flow due to any flow interruption. The well also requires frequent wireline run to make sure tubing walls are clear.

The novel rigless ESP deployment at this well was executed using a structured three-run sequence. The first run involved slickline deployment to set a 3.5″ SIM Plus packer, initially planned at 2622 ft but adjusted to 2580 ft due to tool restriction issues at SPM#1. The second run used E-Lin to latch onto the packer and initiate the kickstart operation with the novel pump, designed to unload heavy fluids and rejuvenate natural flow. Once flow was verified, the third run retrieved the packer, concluding the job.

The well sustained production for one-month post-operation, with no HSE or service quality incidents reported. The job was executed safely with full compliance to high-voltage handling protocols and shift-based safety briefings. The pump was deployed using a electric line cable and operated with a surface VFD unit. The operation achieved successful fluid unloading and natural flow initiation, validating the technology's effectiveness in vertical well section with 3.5″ tubing.

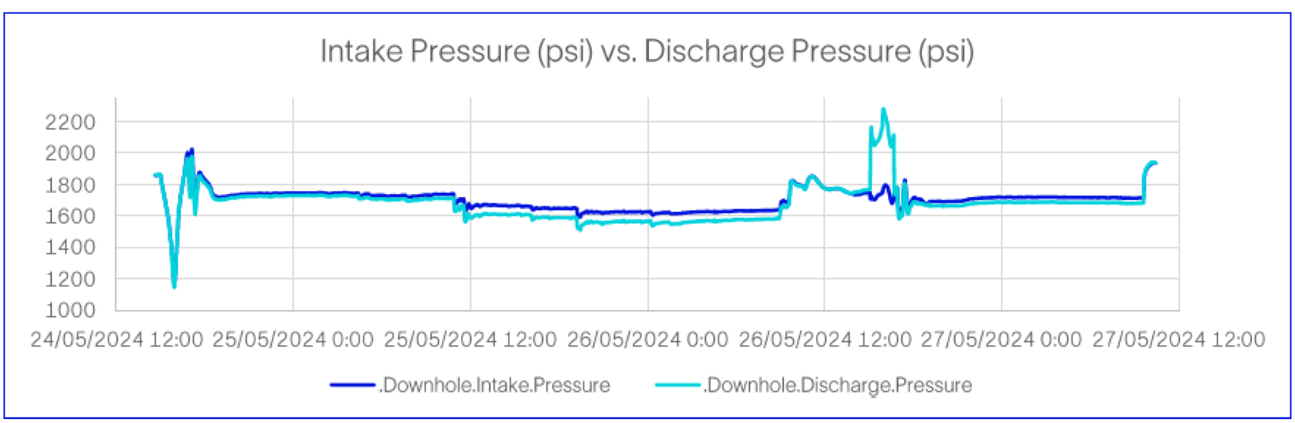

Figure 2—Novel Pump Intake & Discharge Pressure for Well B

Due to higher water cut, well was difficult to be put on production. Pump ran for approximately 3 hours during kick off. Lessons learned from this well included the importance of conducting dummy runs (or Multi Finger Caliper even better) across completion restrictions and keeping the well open prior to intervention to reduce downhole time and improve performance. The trial demonstrated this novel pump's capability to deliver rigless artificial lift with minimal footprint, reduced crew size, and operational flexibility, positioning it as a viable alternative to conventional CT assisted Nitrogen kick off.

Case Study: Well C, Field Y (Oil Producer with horizontal open hole completion & GLMs)

This prospective Well C was located in another field Y. It was completed in 2016 as single oil horizontal open hole producer with 3 ½"completion and 4 SPMs. Just as the previous two wells, this well was also sensitive to flow but not equipped with gas lift yet. The well was delivering liquid in the order of couple of thousand bbl liquid per day with water cut varying between 25 to 30%. GOR ranges up to 1000 scf/bbl.

The job was executed using a **three-stage wireline deployment strategy**. The first stage involved setting a **3.5″ SIM Plus packer** at a depth of **3677 ft**, followed by the second stage where the novel pump was conveyed via **Electric Line** to initiate the kickstart process. This phase focused on unloading wellbore fluids, with the pump running for approximately **7 hours**. The final stage involved retrieving the packer after confirming flow initiation.

Operational modelling prior to deployment predicted a kickstart time of **0.74 hours**, but actual performance extended to **2–2.7 hours**, reflecting the well's marginal inflow characteristics. The well achieved a **flow rate of almost thousand BLPD**, though sustained production lasted only **one day**, indicating limited reservoir support or fluid mobility. Despite this, the intervention was completed without any HSE or service quality incidents, showcasing the safety and efficiency of the novel rigless ESP system.

The trial highlighted several technical learnings: choke size planning was essential to optimize flow post-pump retrieval, and pre-job dummy runs across completion restrictions were recommended to avoid tool hang-ups. The operation also demonstrated the value of maintaining well openness prior to intervention to reduce downhole time and improve fluid dynamics.

Overall, deployment in this well validated the feasibility of rigless ESP technology, offering a lower-cost, faster alternative to conventional nitrogen kick-off methods. While the flow sustainability was short-lived, the job reinforced the adaptability of wireline-deployed ESPs for selective well reactivation and provided critical data for future optimization strategies.

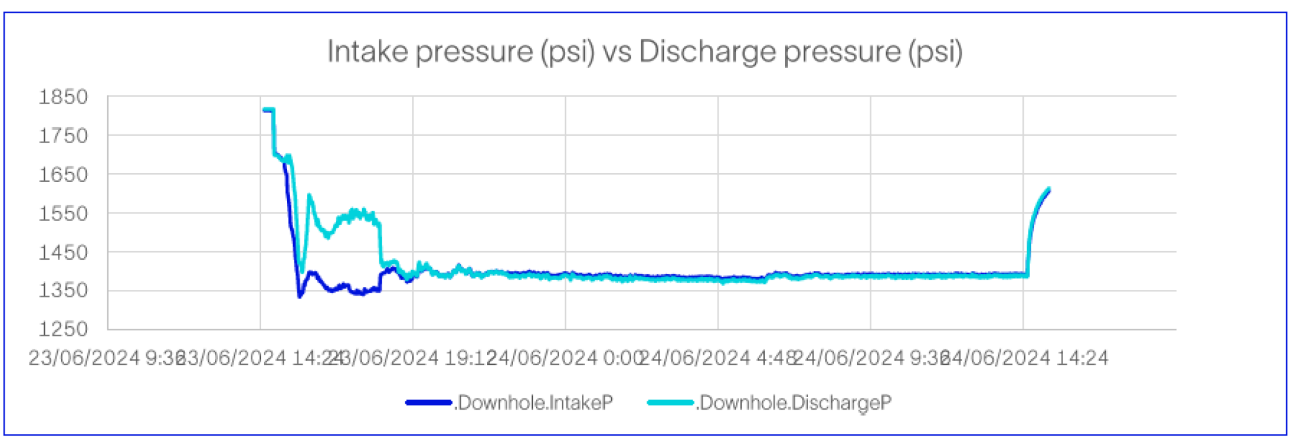

Figure 3—Novel Pump Intake & Discharge Pressure for Well C

Case Study: Well D, Field Y (Oil Producer with horizontal open hole completion & GLMs)

Well D was a single oil producer completed with a 3-1/2″ VAMTOP Cr-13 tubing string extending to a total measured depth of ~**12,000 ft** and a true vertical depth of ~**8,000 ft**. The well featured multiple side pocket mandrels (SPMs), installed with dummy valves and includes a 7″ retrievable packer set at ~**7,700 ft**. The tubing string comprised various pup joints and flow couplings, with a travel joint stroke of 5 ft and a WLEG shoe at ~**9,400 ft**. The casing program includes 13-3/8″ casing set at ~**1,750 ft**, 9-5/8″ casing to ~**6,800 ft**, and 7″ casing reaching ~**9,500 ft**, all cemented with silica blend and neat slurries. The completion design supports artificial lift via gas lift mandrels and is configured for high-pressure production with a 5,000 psi-rated Christmas tree and tubing head. The well produces

Well was completed in 2017 and flowed with almost no water. However, since past two years, it had started flowing with up to 25% water. With the increased water cut, well became sensitive to flow. It required nitrogen kick off using coil tubing whenever ceased to flow. Due to gas lift facility limitations, coil tubing was so far the only option. The well produced approximately couple of thousand bbl liquid per day with 25% water.

The novel technology was attempted on this well. The operation involved a coordinated sequence of activities: initial deployment of a 3.5″ SIM Plus packer at ~3,700 ft, followed by the descent of the novel pump via E-Line to initiate fluid displacement and activate flow. After verifying flow response, the packer was extracted to conclude the job. This well required the longest active pumping duration of the trial—22 hours and 40 minutes—due to complex downhole conditions and fluid behaviour.

Despite the extended effort, the well sustained production for only one day, underscoring its sensitivity to inflow and reservoir support. The operation was executed without any HSE or service quality incidents, and the pump was retrieved successfully. However, the team encountered motor current interruptions attributed to scale buildup and high-water content, which demanded real-time monitoring and operational adjustments.

Post-job analysis emphasized several technical recommendations: integrating protective screens in the bottom hole assembly to shield the pump from debris, and applying incremental speed modulation when intake pressure fails to exceed discharge pressure—allowing the well to transition into natural flow. Additionally, the importance of choke size planning was highlighted to optimize both unloading and post-intervention flow stabilization 1.

While the production outcome was limited, the SB-779 operation validated the rigless deployment and retrieval capabilities of novel technology. It demonstrated the system's adaptability under high water cut conditions and reinforced its potential as a rapid, low-footprint alternative to conventional nitrogen-based kick-off methods.

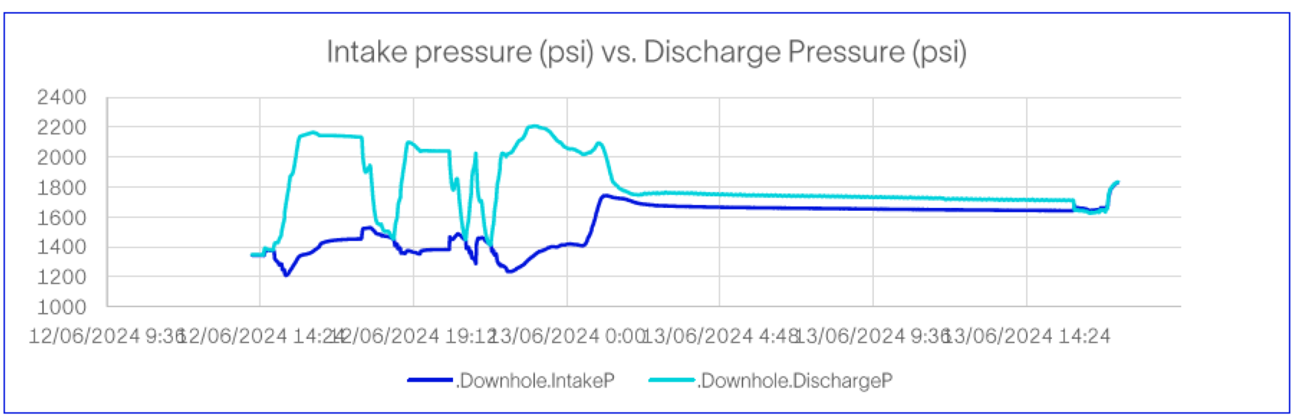

Figure 4—Novel Pump Intake & Discharge Pressure for Well D

Sustainability Impact

Well activation is a critical operation in oilfield production, and selecting the right method can significantly impact operational efficiency, cost, and environmental footprint. This analysis compares two activation methods—Novel electric submersible lift system and Nitrogen Kick-Off using Coiled Tubing (CT)—based on following assumed diesel consumption and associated CO_2 emissions.

Category	Parameter	Novel ESP	Nitrogen Kick-Off (CT)
Mobilization Distance	From base to wellsite	40 km	40 km
Transport Vehicles	Trucks, Crane	2 Trucks + 1 Crane	4 Trucks + 1 Crane
Diesel for Mobilization	Liters	48 L	80 L
	CO_2 Emissions	128.64 kg	214.40 kg
Rig-Up Duration	Setup time	1 day	4 days
Rig-Up Diesel Consumption	Liters	100 L	720 L
	CO_2 Emissions	268.00 kg	1,929.60 kg
Operation Duration	Activation time	3 hours	8 hours
Diesel for Activation	Liters	120 L	1,040 L
	CO_2 Emissions	321.60 kg	2,787.20 kg
Total Diesel Consumed	Liters	**268 L**	**1,840 L**
Total CO_2 Emissions	Kilograms	**718.24 kg**	**4,931.20 kg**

Following the successful deployment of this novel rigless ESP for well activation, as per the assumptions in the table above, this operation demonstrated clear advantages in terms of efficiency, environmental impact, and logistical simplicity. With a total diesel consumption of just **268 liters** and associated **CO_2 emissions of 718.24 kg**, this new technology significantly outperformed the traditional nitrogen kick-off method, which consumed **1,840 liters** and emitted **4,931.20 kg** of CO_2. The reduced rig-up time (1 day vs. 4 days), lower fuel usage, and minimal equipment footprint contributed to a faster, safer, and more sustainable activation process. These results reinforce this novel rigless ESP's suitability for future well interventions, particularly in operations where environmental performance and cost-efficiency are key priorities.

Conclusion

The deployment of novel rigless ESP technology across fields marked a significant milestone in the evolution of sustainable well activation strategies. Designed for rigless, through-tubing deployment, the system successfully reactivated ceased wells with minimal surface footprint, reduced operational complexity, and enhanced safety. Compared to conventional coiled tubing nitrogen kick-off, the technology demonstrated an 85% reduction in diesel consumption and CO_2 emissions, across mobilization, rig-up, and execution phases.

Its compact design enabled access to slim-bore and deviated wells, while eliminating the need for rig interventions lowered lifecycle costs and reduced HSE exposure. The system performed reliably in varied reservoir conditions, maintaining flow in sartificial lift scenarios and proving its adaptability to complex well architectures.

This trial not only validates the technology's operational and environmental benefits but also aligns with industry's sustainability objectives and the UAE's broader decarbonization goals. It offers a scalable, efficient, and eco-friendly alternative to traditional activation methods, positioning itself as a transformative solution for future well interventions in mature and challenging fields.

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